Essays

|!|BCEEB0260000001|!| ★ **Cross topic**

Discuss the role of genetic and environmental factors in determining the colour of human skin. (10 marks)

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Genetic Factors: (Max. 5)

- Skin colour is controlled by genes that regulate the production of melanin. (1)
- Different combinations of the alleles for such genes between individuals causes variation in skin colour. (1)
- During meiosis, the independent assortment of chromosomes leads to the formation of gametes with different combinations of alleles that control melanin production. (1)
- During meiosis, crossing over also produces gametes with new combinations of alleles that control melanin production. (1)
- During fertilization, male and female gametes with different genetic makeup fuse together randomly. This gives rise to children with different combinations of alleles for melanin production and thus leads to different skin colour. (1)
- If mutations occur in the genes that control the production of melanin, the structure or function of the proteins controlled by these genes may be altered. These could lead to changes in skin colour. (1)

Environmental Factors: (Max. 2)

- Environmental factors can also affect the production of melanin in humans. (1)
- Frequent exposure to sunlight causes the human body to produce more melanin, which results in darker skin. (1)
- Low exposure to sunlight causes the human body to produce fewer melanin, which results in lighter skin. (1)

(Other reasonable answers)

Communication (Max. 3)

(10 marks)

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Skills assessed	Marks
Applying concepts	7
Communication	3

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